

Cross-column RT calibration report

METHOD	cross-column calibration onto a reference column (v2)
REFERENCE COLUMN	Column 25 (bundled default)
STANDARDS PANEL	mix15
STAGE 1 CURVE	595 m/z-matched pairs (173 standards-run + 422 sample-run), 421 kept after outlier trimming
STAGE 2 ANCHORS	correction APPLIED (lam_g=0.6, lam_c=0.5) - anchor RMS 0.206 -> 0.174 min leave-one-out
IRT SCALE	1-100, affine on the reference RT axis between the earliest and latest of 14 landmarks (1.46-21.61 min); landmark panel mix15

FEATURES

samples	22,314
standards run	8,768
feature rows in each table	

MATCHED PAIRS

standards	173
sample	422
421 of 595 kept after MAD trimming	

CURVE RESIDUAL

median	0.366
P90	4.213
reference – fitted (min)	

ANCHOR GATE

on · 29%

lam_g 0.6 · lam_c 0.5 · 15 anchors

IRT LANDMARKS

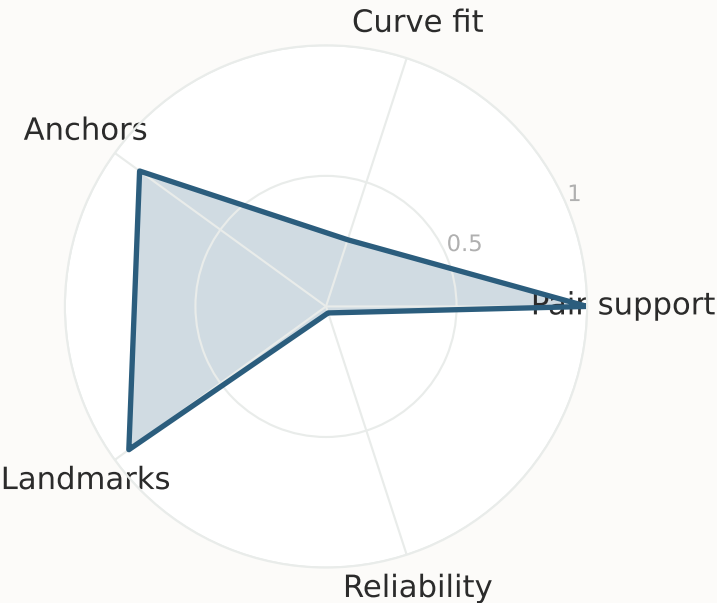
14

RT 1.46-21.61 min on the reference run

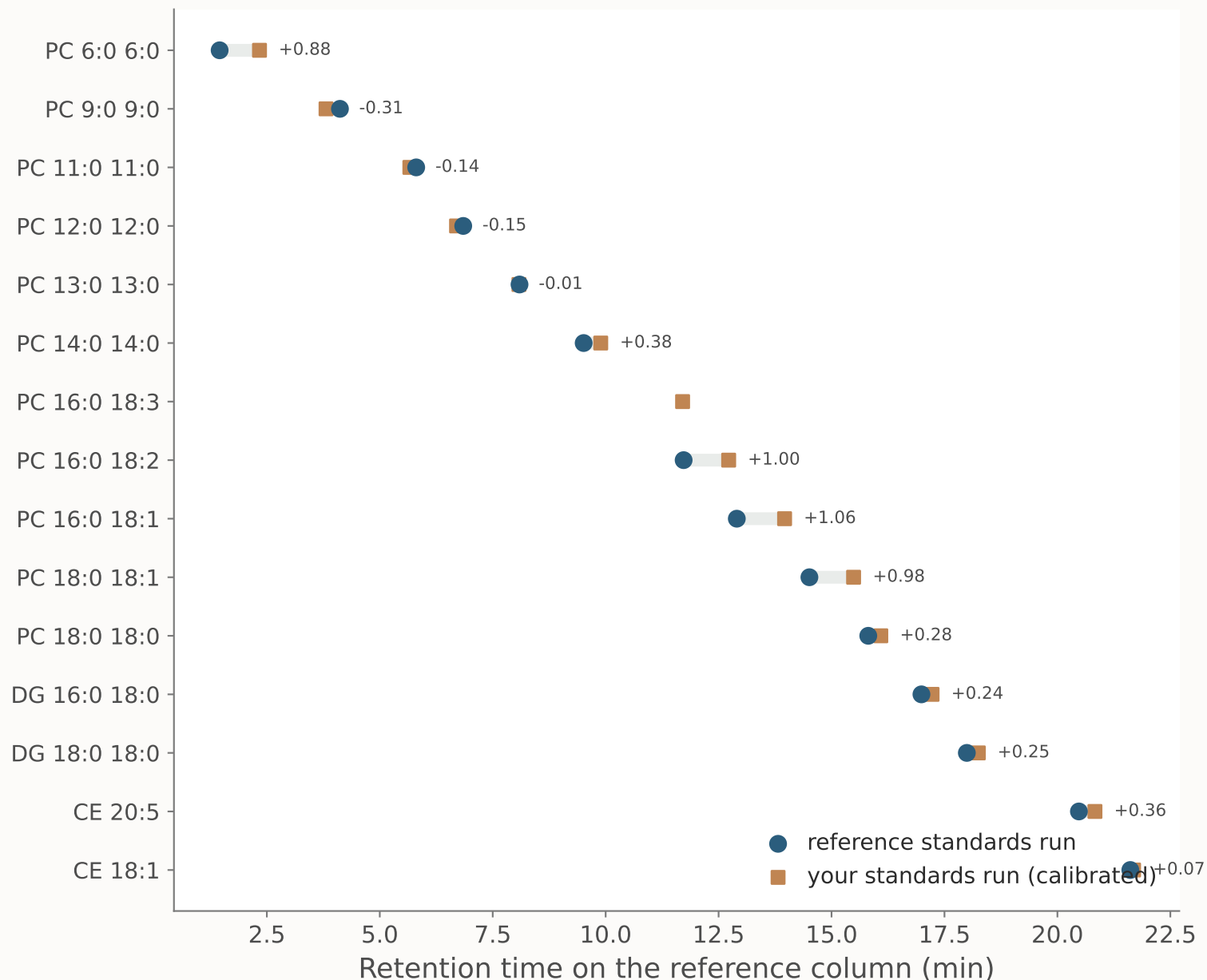
OUTPUT

iRT range	-5-100
extrapolated	2.5%
3% high-reliability	

Quality fingerprint

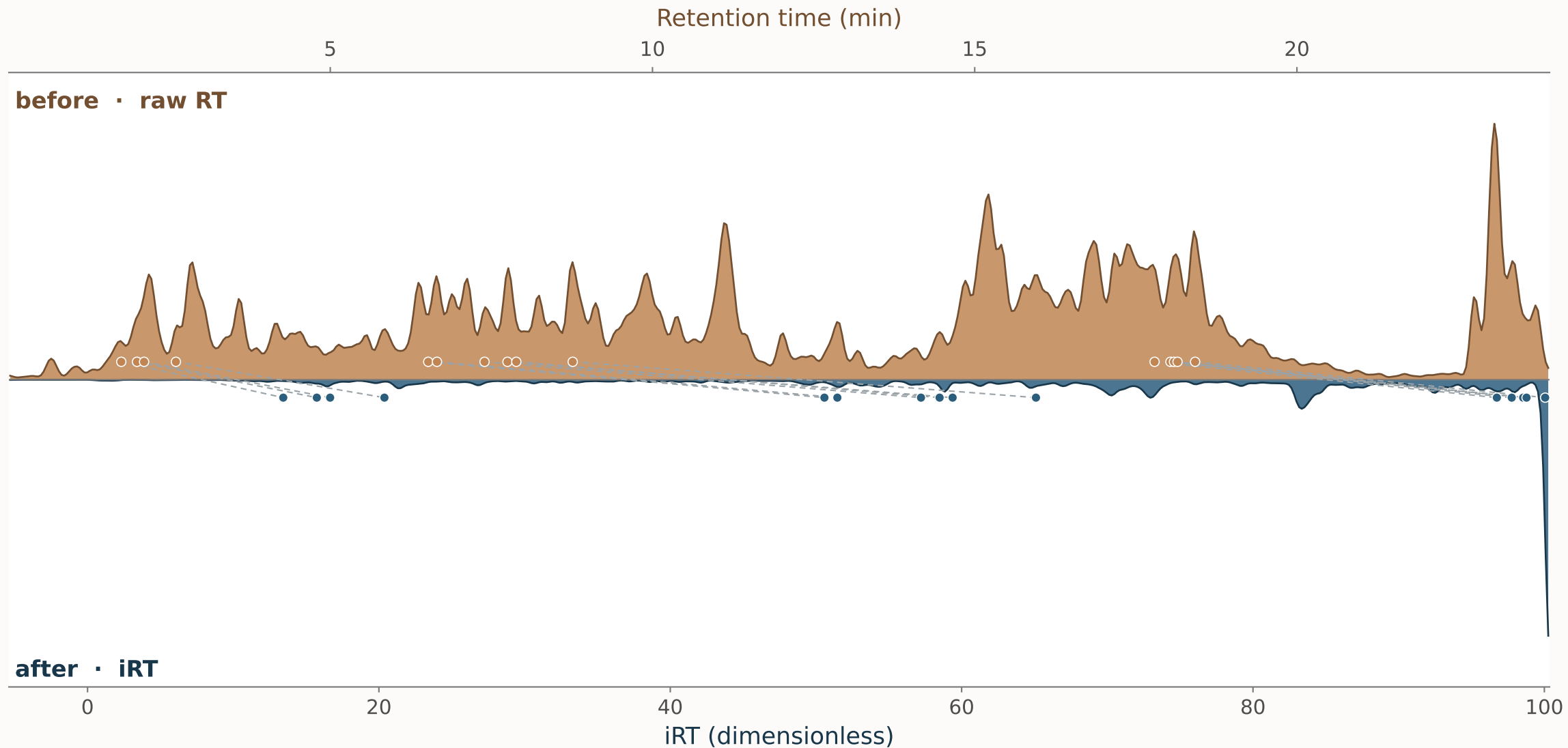


Standard detection — reference vs panel vs samples



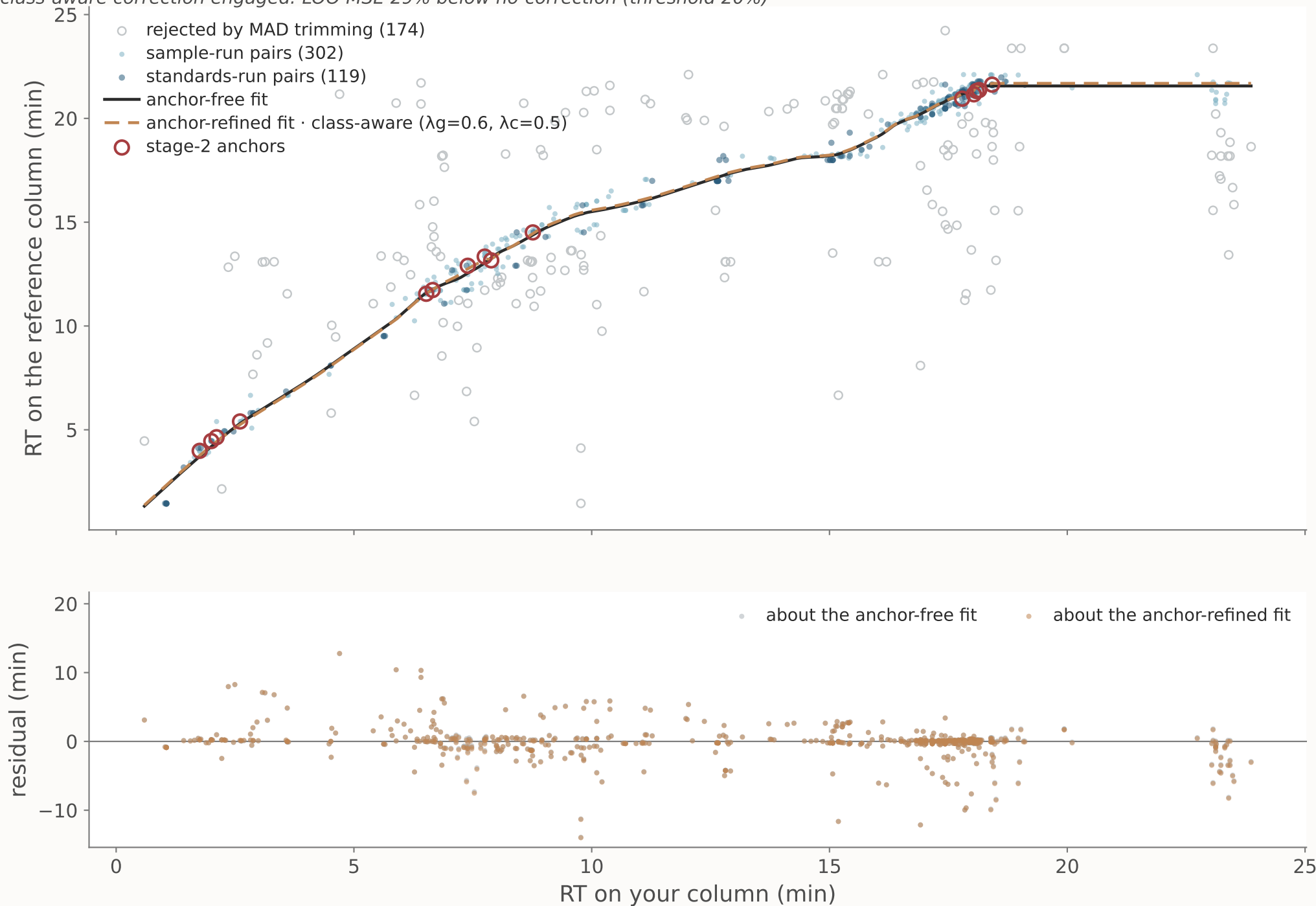
Reconstructed feature-intensity profile · before vs after calibration

stage-2 anchors connected — line slant = RT offset · reconstructed from the feature table (binned intensity)



Cross-column calibration curve · your column -> the reference column

class-aware correction engaged: LOO MSE 29% below no-correction (threshold 20%)



Stage-2 sample anchors

correction APPLIED ($\text{lam}_g=0.6$, $\text{lam}_c=0.5$) - anchor RMS 0.206 -> 0.174 min leave-one-out

class median offsets (min): CE -0.092 LPC +0.245 PC +0.049 TG -0.197

2 anchor(s) removed by the sanity filter

